DEPI PROJECT

NUVANTA

AI-Powered Enterprise Financial & Operations Management System

Project Plan – Timeline

Document	Project Plan – Timeline (Gantt, Milestones, Deliverables, Resources)
Project	NUVANTA
Instructor	Ahmed Mohamed Maher
Track	Full-Stack .Net Web
Team	Menna Atef, Nour Hany, Weam Emad, Nourhan Fawzy, Omar Mohamed

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1. Timeline Overview

1.1 Purpose

This document turns the NUVANTA Project Proposal into an executable schedule. It shows what is built each week, who works on it, which milestones mark progress, and which deliverables the team hands in. It covers **Phase 1 (Core MVP)** and the three selected AI features only; Phases 2–4 of the proposal stay as future work.

1.2 Duration and Planning Assumptions

- The plan runs for **6 provisional project weeks** (Week 1 to Week 6). Calendar dates are still to be confirmed and will be mapped to the official DEPI deadlines.
- **Scope freeze at the end of Week 1** (risk R1): only Phase 1 items listed in the proposal are built.
- Work is organised as **vertical slices** so the full flow (budget > purchase request > approval > receipt > payment > effect on budget, ledger and inventory) works as early as possible.
- **Feature freeze at the end of Week 5.** Week 6 is used for integration testing, bug fixing, documentation and the demo, and acts as the schedule buffer.
- Permission and financial-integrity tests are written alongside each module, not left to the end (risks R2–R5).
- Team size is 5 members; everyone contributes to documentation, code reviews and testing, as in the Ways of Working.

1.3 Phases at a Glance

Phase	Weeks	Focus	Exit Milestone
A. Initiation & Design	W1–W2	Final SRD, scope freeze, architecture, database schema, OpenAPI contracts, wireframes, repo, Docker and CI/CD	M1, M2
B. Core Build	W2–W4	Users and RBAC, budgets, purchase requests, approvals, suppliers, POs, goods receipt, invoices	M2, M3
C. Finance, Inventory & Integration	W4–W5	Payments (sandbox and external), cash, ledger, inventory, audit logs, notifications, dashboards	M4
D. AI Layer	W2–W6	Seed data, cash-flow and budget forecasting, anomaly detection, AI Business Assistant	M5
E. Testing & Delivery	W3–W6	Permission and financial tests, integration testing, bug fixing, final documentation, demo	M6

2. Gantt Chart

Bars show the weeks in which a task is active. Owners: **MA** Menna Atef, **NF** Nourhan Fawzy, **NH** Nour Hany, **WE** Weam Emad, **OM** Omar Mohamed (first name is the lead; all tasks are reviewed through pull requests).

#	Task	Owners	W1	W2	W3	W4	W5	W6
A. Planning & Design								
1	Finalize SRD and freeze scope	MA, all	■					
2	Architecture, DB schema, OpenAPI contracts	MA, OM, NH	■	■				
3	UI wireframes and design system	NF, WE	■	■				
4	Repo, Jira, Docker Compose, CI/CD	OM	■	■				
B. Core Backend								
5	Auth, Users, Roles, Permissions, Departments	OM, NH		■	■			
6	Budget setup and allocation	MA		■	■			
7	Purchase requests and approval workflow (Maker–Checker)	MA, NH			■	■		
8	Suppliers, POs, goods receipt, supplier invoices	WE			■	■		
9	Payments: sandbox and external transfer recording	WE, OM				■	■	
10	Cash transactions, budget tracking, basic ledger	MA, NH				■	■	
11	Basic inventory	WE				■	■	
12	Audit logs and notifications	OM, WE				■	■	
C. Frontend								
13	React app shell, login, navigation	NF		■	■			
14	Budget, purchase request, approval inbox screens	NF, WE			■	■	■	
15	Role-based dashboards and responsive polish	NF				■	■	■
D. AI Layer								
16	Realistic seed / synthetic data	NH		■	■			
17	Cash-flow and budget forecasting	NH, MA				■	■	
18	Anomaly detection (rules first, then Isolation Forest)	NH				■	■	
19	AI Business Assistant (permission-checked tools)	MA, NH					■	■
E. Testing & Delivery								
20	Permission and financial-integrity tests	OM, all			▒	▒	▒	▒
21	Integration testing and bug fixing	All					■	■
22	Final documentation, demo rehearsal, presentation	All						■
F. Project Management								
23	Weekly meetings, Jira and PR reviews	MA	▒	▒	▒	▒	▒	▒
	Milestones (end of week)		**M1**	**M2**	**M3**	**M4**	**M5**	**M6**

■ Active task ▒ Continuous activity ■ Milestone

3. Milestones

Each milestone has a clear check so the team can tell at the weekly review whether the project is on schedule.

ID	Due	Milestone	Completion Check
M1	End of W1	Scope frozen and foundations ready	SRD and scope approved by the team lead; architecture, DB schema and API contracts drafted; repo, Jira board and Docker Compose running
M2	End of W2	Security and budget foundations	Login, roles, permissions and departments work with backend checks; budget allocation works; CI pipeline runs on every pull request
M3	End of W3	Request-to-order flow	Purchase request is created, checked against available budget, approved by a different user (Maker–Checker) and turned into a purchase order
M4	End of W4	End-to-end MVP workflow	Full demo scenario passes: budget > request > approval > receipt > payment > updated budget, ledger and inventory; failed and repeated payments handled
M5	End of W5	Feature freeze with AI integrated	Forecasting, anomaly detection and AI Assistant connected to the backend; dashboards, audit logs and notifications done; permission tests pass
M6	End of W6	Final delivery and demo	Zero critical bugs; test targets met; documentation complete; demo rehearsed; all DEPI deliverables submitted

4. Deliverables

ID	Deliverable	Week	Lead	Acceptance Criteria
D1	Final SRD with numbered requirements	W1	MA	Reviewed by the team lead; scope frozen
D2	Architecture, ERD and OpenAPI specification	W1–W2	MA, OM	Agreed by all members; used by frontend and backend
D3	UI wireframes and role-based dashboard designs	W2	NF, WE	Cover the main screens of the demo scenario
D4	Working environment: GitHub repo, Jira, Docker Compose, CI/CD	W2	OM	Full stack starts with one command; CI runs on each PR
D5	Security and budget module	W3	OM, MA	Role and data-scope tests pass; allocations never exceed the total budget
D6	Procurement module (requests, approvals, suppliers, POs, receipt, invoices)	W4	WE, MA	Maker–Checker enforced; full request history kept
D7	Payments, cash, ledger and inventory modules	W5	WE, MA, OM	No duplicate payments; budget, ledger and stock update correctly
D8	AI service: forecasting, anomaly detection, AI Assistant	W5–W6	NH, MA	MAPE and precision/recall reported; Assistant respects permissions
D9	Test suite and traceability matrix	W6	OM, all	At least 90% of requirements covered; pass rate at least 95%
D10	Final documentation, demo and presentation	W6	All, MA reviews	Demo checklist passes; submitted on time

5. Resource Allocation

5.1 Team and Main Focus

Roles follow the Task Assignment section of the Project Planning & Management document. Members work across modules and review each other's pull requests, so the main focus below is a guide, not exclusive ownership.

Member	Role	Main Focus in this Plan
Menna Atef	Team Lead, Backend & AI Developer	Scope and SRD, finance core, budget, approval workflow, ledger, AI integration; reviews all deliverables
Nourhan Fawzy	Frontend & Backend Developer	Wireframes, React shell, budget and request screens, dashboards, responsive design
Nour Hany	Backend & AI Developer	Seed data, users and approval APIs, financial data processing, forecasting, anomaly detection
Weam Emad	Frontend & Backend Developer	Suppliers, POs, goods receipt, payments, inventory, approval inbox and notification screens
Omar Mohamed	Backend & DevOps Engineer	Repo, Docker, CI/CD, auth and RBAC, audit logs, automated tests, deployment

5.2 Weekly Allocation

Member	W1	W2	W3	W4	W5	W6
Menna Atef	SRD, scope freeze, architecture	Budget module	Approval workflow	Ledger, budget tracking	AI integration, forecasting review	Final review, demo
Nourhan Fawzy	Wireframes, design system	React shell, login	Budget and request screens	Dashboards	Responsive polish, UI fixes	UI bug fixes, demo
Nour Hany	API contracts, data model	Seed data, users and department APIs	Request and approval APIs	Forecasting model	Anomaly detection, Assistant tools	AI evaluation, fixes
Weam Emad	Wireframes, supplier data model	Supplier APIs and UI	POs, goods receipt, invoices	Payments, inventory	Approval inbox, notifications UI	UI testing, docs
Omar Mohamed	Repo, Docker, CI/CD	Auth, RBAC	Test framework, permission tests	Audit logs, payment idempotency	Financial tests, deployment	Release, regression tests

5.3 Tools and Environment

Resource	Use
Jira	Tasks with descriptions, priorities, acceptance criteria and due dates; board updated by every member
GitHub	Feature branches, meaningful commits, pull requests with at least one reviewer, CI results
Docker Compose	Runs the .NET backend, SQL Server, Python AI service and React frontend together
Stack	React, TypeScript, Vite, Tailwind; ASP.NET Core Web API, EF Core, SQL Server; Python, FastAPI, scikit-learn
Meetings	Weekly team meeting and a quick check-in before each DEPI deadline

5.4 Capacity Risks

- Task owners above are planned leads; if a member is unavailable (risk R10), pair on the critical modules and redistribute at the weekly meeting.
- Weeks 4 and 5 are the busiest. If either slips, move Phase D items (anomaly detection or Assistant) first, never the end-to-end MVP workflow.